Gaming Castle
Online Gaming Centre Management System

Software Requirements Specification (SRS)

Version 03

By

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Version History

Version No.	Date	Author(s)	Summary of Changes
01	2026/01/15	Kirisigan, Jashvanth, Thivya, Nirojan	Initial draft of SRS document
02	2026/03/01	Kirisigan, Jashvanth, Thivya, Nirojan	Corrections applied: figure/table captions added, ToC updated, heading format corrected, Stakeholder typo fixed, passive voice revisions, IEEE reference format corrected
03	2026/10/22	Kirisigan, Jashvanth, Thivya, Nirojan	Aligned SRS with the implemented system: multi-category stations (PS5, PC Esports, VR); real-time 5-minute slot hold; cancellation/rescheduling policy; session ratings and reviews; PayHere (LKR) payments with IPN webhook and card validation; loyalty reversal and audit trail; Saga compensation and circuit breakers; gateway rate limiting; six-service architecture with a Socket.IO real-time server; new FR-32 to FR-48, NFR-19 to NFR-24, BR-09 to BR-15, UC-15 and UC-16; updated risks, data requirements and references

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Chapter 1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document is produced to define the functional and non-functional requirements for the Gaming Castle Online Gaming Centre Management System. A clear and comprehensive description of the system to be developed is provided by this document, and it is intended to serve as a contractual agreement between the development team and the client.

The intended audience of this document is identified as the project development team, academic evaluators, and the client (Gaming Castle management). The design, development, testing, and deployment phases of the system are guided by this document. Version 03 reflects the system as implemented on the current development branch and supersedes versions 01 and 02.

1.2 Scope

The system to be developed is named Gaming Castle, and it is designed as an Online Gaming Centre Management Web Application to support the operations of the Gaming Castle gaming hub located in Nelliyadi, Sri Lanka.

An integrated platform for managing gaming station bookings, tournament registrations, payment processing, customer loyalty programs, session reviews, and administrative operations shall be provided by the system. Three categories of gaming station shall be managed: PlayStation 5 (PS5) console stations, PC Esports rigs, and Virtual Reality (VR) stations. Registered customers shall be allowed to book slots online, hold a slot during checkout, pay online in Sri Lankan Rupees (LKR), take part in tournaments, redeem loyalty points, and review completed sessions.

Two primary user roles, namely Customer and Administrator, shall be supported by the system. Registration, login, gaming slot reservations, cancellation and rescheduling, tournament participation, session reviews, and the viewing of booking and reward history shall be enabled for customers. Booking management (including walk-in bookings with cash payment), station and game management, tournament organisation and supervision, activity monitoring, and report generation for decision-making shall be carried out by administrators.

The system shall be developed as a web-based application, and accessibility through modern web browsers shall be ensured for both customers and administrators.

The development of games, game streaming services, integration with external gaming platforms such as the PlayStation Network, and hardware-level control of consoles, PCs, or VR headsets shall not be included in the system.

1.3 Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification

Term / Acronym	Definition
FR	Functional Requirement
NFR	Non-Functional Requirement
BR	Business Rule
Admin	System Administrator – Gaming Castle staff managing the platform
User / Customer	Registered individual who books slots, participates in tournaments, or reviews sessions
Station	A bookable gaming unit of type Console (PS5), PC (Esports rig), or VR
Slot	A reserved time block (minimum one hour) on a station at Gaming Castle
Soft Lock / Slot Hold	A temporary 5-minute reservation of a slot during checkout, enforced by the real-time server
TTL	Time To Live – the duration after which a soft lock expires automatically
Tournament	A competitive gaming event organised and managed through the system
Loyalty Program	A rewards scheme in which points are earned on spend and redeemed as discounts
Review	A 1–5 star rating with optional comment submitted for a completed booking
PayHere	Sri Lankan payment gateway used for LKR online payments
IPN	Instant Payment Notification – asynchronous webhook sent by PayHere to confirm a payment
Luhn algorithm	Mod-10 checksum used to validate payment card numbers on the client
Saga	A pattern that maintains data consistency across services using local transactions with compensating actions
Compensating Transaction	A reverse action (for example, loyalty point reversal or slot release) executed when a later Saga step fails
Circuit Breaker	A resilience pattern that stops calls to a failing service and returns a fallback response
Token Bucket	A rate-limiting algorithm that permits bursts up to a capacity and refills at a fixed rate
WebSocket / Socket.IO	Bidirectional real-time communication protocol and library used for live slot and notification updates
LKR	Sri Lankan Rupee
PS5 / VR	PlayStation 5 / Virtual Reality
UI	User Interface
API	Application Programming Interface
IEEE	Institute of Electrical and Electronics Engineers
ISO/IEC 25010	International standard for software product quality characteristics
MVP	Minimum Viable Product
UAT	User Acceptance Testing

Table 2: Definitions, Acronyms and Abbreviations

1.4 Document Overview

This document is organised as follows:

- Chapter 1 – Introduction: Provides the purpose, scope, definitions, and overview of the document.
- Chapter 2 – Project Scope and Stakeholders: Describes project boundaries, station categories, and identifies all key stakeholders.
- Chapter 3 – Functional Requirements: Lists all functional requirements grouped by system modules in tabular format.
- Chapter 4 – Non-Functional Requirements: Details measurable quality requirements categorised per ISO/IEC 25010.
- Chapter 5 – Business Rules, Constraints, and Assumptions: Documents operational rules and project limitations.
- Chapter 6 – Use Case Diagrams and Descriptions: Presents use cases for all system actors.
- Chapter 7 – Activity Diagrams: Illustrates key workflows within the system.
- Chapter 8 – External Interface Requirements: Specifies UI, hardware, software, and communication interfaces.
- Chapter 9 – Data Requirements: Defines the data entities and storage requirements.
- Chapter 10 – Alternative Solutions Considered: Discusses alternative approaches evaluated.
- Chapter 11 – Feasibility Study: Assesses technical, economic, operational, and schedule feasibility.
- Chapter 12 – Risk Analysis: Identifies and evaluates project risks with mitigation strategies.
- Chapter 13 – References: Lists all cited sources and standards.

Chapter 2 Project Scope and Stakeholders

2.1 Project Scope

Gaming Castle is a modern gaming hub located in Nelliyadi, Sri Lanka, where PS5 console stations, PC Esports rigs, and VR stations are operated and a selection of popular games is offered on a pay-as-you-go basis. Manual processes were previously relied upon for booking, payments, tournament management, and customer offers, resulting in operational inefficiencies.

All key operations of Gaming Castle shall be digitised and streamlined by the proposed web-based system. Real-time multi-category station booking with slot holding, PayHere online payments, structured tournament registration, a loyalty discount programme, a session review system, real-time notifications, and a comprehensive administrative dashboard shall be enabled by the system.

2.2 Supported Station Categories

Category	Station Type	Equipment	Hourly Rate
PS5 Lounge	CONSOLE	PS5 with DualSense wireless controllers, 4K 120 Hz display	LKR 600
PC Esports Rig	PC	RTX 4080 GPU, mechanical keyboard, 240 Hz monitor	LKR 700
VR Rig	VR	Meta Quest Pro / PSVR2, 360-degree tracking	LKR 800

Table 3: Supported Station Categories

Hourly rates are stored per station and may be changed by the administrator; the values above are the rates at the time of writing.

2.3 Stakeholders

Stakeholder	Role	Interest in System
Gaming Castle Owner	Client / Primary Stakeholder	Oversees overall business operations, monitors revenue and station utilisation, and ensures business growth and efficiency through the system.
Administrator (Staff)	System User – Admin Role	Manages bookings, stations, games, tournaments, payments, user accounts, and reports.
Registered Customers	System User – Customer Role	Books and manages slots, joins tournaments, earns loyalty rewards, and reviews sessions.
Walk-in Customers	Indirect User	Uses the system indirectly through administrator-assisted booking of gaming slots.
Development Team	System Developers	Responsible for designing, developing, testing, and deploying the system according to requirements.
Project Supervisor	Academic Stakeholder	Provides academic oversight and evaluation.
University of Kelaniya	Academic Institution	Sets degree requirements and quality standards.

Stakeholder	Role	Interest in System
PayHere	External System	Processes LKR card and wallet payments and sends IPN webhooks.
Email / SMS Providers	External Systems	Deliver email (SMTP) and SMS (Twilio) notifications and verification codes.

Table 4: Stakeholders

Chapter 3 Functional Requirements

The Change column shows how each requirement differs from SRS version 02: Retained, Modified, or New. Requirement IDs from version 02 are preserved to keep traceability; new requirements continue from FR-32.

3.1 User Registration and Authentication

ID	Description	Priority	Stakeholder	Change
FR-01	The system shall allow customers to register using their full name, email address, phone number, and password. Passwords shall be stored as bcrypt hashes.	High	Customer	Modified
FR-02	The system shall allow registered users to log in using their email and password.	High	Customer	Retained
FR-03	The system shall allow registered users to log in using their phone number and password.	High	Customer	Retained
FR-04	The system shall allow users to reset their passwords using a time-limited one-time verification code sent by email. Repeated failed code attempts shall lock the reset attempt.	High	Customer	Modified
FR-05	The system shall allow users to reset their passwords using a time-limited one-time verification code sent by SMS to the registered phone number, with the same failed-attempt lockout.	High	Customer	Modified
FR-06	The system shall allow the administrator to manage user accounts: search, view profile and booking history, update details, and activate or deactivate accounts.	High	Admin	Modified
FR-32	The system shall allow customers to deactivate their own accounts and to reactivate them through an email-verified workflow.	Medium	Customer	New
FR-33	The system shall allow customers to manage their profile, upload an avatar image, and view their loyalty tier.	Medium	Customer	New

Table 5: Functional Requirements – User Registration and Authentication

3.2 Station and Slot Booking Management

ID	Description	Priority	Stakeholder	Change
FR-07	The system shall display slot availability in real time by date and by station category (PS5, PC, VR), together with the games available on each station.	High	Customer	Modified
FR-08	The system shall allow registered customers to book one or more consecutive one-hour slots on a station by selecting a date, time range, station, and a compatible active game, and shall compute the total price dynamically from the station's hourly rate.	High	Customer	Modified

ID	Description	Priority	Stakeholder	Change
FR-09	The system shall allow the administrator to book slots on behalf of walk-in customers and to chain cash payment recording with instant booking confirmation.	High	Admin	Modified
FR-10	The system shall prevent double-booking of the same station slot at both the real-time lock layer and the database level (unique constraint).	High	Customer	Modified
FR-11	The system shall allow customers to reschedule a confirmed booking to any available slot within 30 days, without fee, up to 2 hours before the scheduled start time.	High	Customer	Modified
FR-12	The system shall send booking confirmation and change notifications to customers via email and SMS.	Medium	Customer	Modified
FR-34	When a customer proceeds to checkout, the system shall soft-lock the selected slot for 5 minutes and broadcast a `slot_locked` event so that all connected clients display a "Holding" indicator; on expiry or payment failure the slot shall be released and a `slot_unlocked` event broadcast.	High	Customer	New
FR-35	The system shall allow customers to cancel a confirmed booking up to 2 hours before its start time, immediately releasing the slot and reversing any loyalty points awarded or redeemed for the booking.	High	Customer	New

Table 6: Functional Requirements – Station and Slot Booking Management

3.3 Session Ratings and Reviews

ID	Description	Priority	Stakeholder	Change
FR-36	The system shall allow customers to rate a completed gaming session from 1 to 5 stars with optional feedback text (for example, comments on the controller or rig).	High	Customer	New
FR-37	The system shall allow only one review per completed booking and shall prevent reviews of cancelled or upcoming bookings.	Medium	Customer	New
FR-38	The system shall aggregate review ratings per station and display the average rating and star badge on station selection cards.	Medium	Customer	New
FR-39	The system shall provide customers with a "My Reviews" history and shall allow administrators to view customer reviews.	Medium	Customer / Admin	New

Table 7: Functional Requirements – Session Ratings and Reviews

3.4 Tournament Management

ID	Description	Priority	Stakeholder	Change
FR-13	The system shall allow the administrator to create, edit, publish, and cancel tournament events with details (game, date, entry fee, maximum participants) and a cover image.	High	Admin	Modified

ID	Description	Priority	Stakeholder	Change
FR-14	The system shall allow registered customers to view upcoming tournaments and register by paying the entry fee; registration shall be refused once the participant cap is reached.	High	Customer	Modified
FR-15	The system shall manage tournament brackets and participant lists.	Medium	Admin	Retained
FR-16	The system shall notify registered participants of tournament schedules and results.	Medium	Customer	Retained
FR-17	The system shall allow the administrator to update tournament status (upcoming, ongoing, completed, cancelled).	Medium	Admin	Modified
FR-18	The system shall include an automated tournament management module that generates brackets and manages match progression based on predefined logic or historical data.	Medium	Admin	Retained
FR-40	The system shall display a real-time tournament ticker banner on customer and admin dashboards showing status, prize pool, and spots remaining.	Medium	Customer / Admin	New
FR-41	When a tournament is cancelled, the system shall notify every registered participant by email and by a real-time browser alert.	Medium	Customer	New

Table 8: Functional Requirements – Tournament Management

3.5 Payment Management

ID	Description	Priority	Stakeholder	Change
FR-19	The system shall calculate charges from booked hours and the hourly rate of the selected station type, less any loyalty discount.	High	Customer	Modified
FR-20	The system shall support online payment for slot bookings and tournament entry fees through the PayHere gateway in LKR, using a server-side MD5 merchant signature.	High	Customer	Modified
FR-21	The system shall generate and display a digital receipt on successful payment, including a printable receipt for cash payments.	High	Customer / Admin	Modified
FR-22	The system shall allow the administrator to record cash payments for walk-in customers.	High	Admin	Retained
FR-23	The system shall provide a payment history log accessible to customers (own payments) and the administrator (all payments).	Medium	Admin / Customer	Retained
FR-42	The system shall verify online payments asynchronously by receiving and validating PayHere IPN webhook notifications (`POST /api/payments/payhere/notify`).	High	System	New

ID	Description	Priority	Stakeholder	Change
FR-43	The client card payment form shall auto-format input (4-digit groups, MM/YY expiry), detect the card brand, and validate the card number with the Luhn (Mod-10) algorithm before submission. Card details shall not be persisted by the system.	High	Customer	New
FR-44	If a downstream step fails after a payment has been captured, the system shall trigger compensation (refund or reversal) through Saga compensating transactions.	Medium	System	New

Table 9: Functional Requirements – Payment Management

3.6 Loyalty Program

ID	Description	Priority	Stakeholder	Change
FR-24	The system shall award loyalty points automatically on booking completion (by default 10% of spend converted to points).	High	Customer	Modified
FR-25	The system shall allow customers to redeem loyalty points as a direct discount at booking checkout.	High	Customer	Modified
FR-26	The system shall display a customer's loyalty balance and tier on their profile.	Medium	Customer	Modified
FR-27	The system shall allow the administrator to configure loyalty earning and redemption rules.	Medium	Admin	Retained
FR-45	The system shall automatically reverse loyalty points awarded or redeemed for a booking when the booking is cancelled.	High	Customer	New
FR-46	The system shall keep an audit trail of loyalty transactions (EARNED, REDEEMED, REVERSED) with a running balance after each transaction.	Medium	Customer / Admin	New

Table 10: Functional Requirements – Loyalty Program

3.7 Administrative Dashboard and Notifications

ID	Description	Priority	Stakeholder	Change
FR-28	The system shall provide a real-time administrative dashboard showing active sessions, today's revenue, station utilisation percentage, upcoming bookings, and active tournaments.	High	Admin	Modified
FR-29	The system shall allow the administrator to generate and export (CSV or PDF) reports on booking revenue, station utilisation, and tournament participation.	High	Admin	Modified
FR-30	The system shall allow the administrator to manage stations (type, hourly rate, image, active status) and the game titles available on each station.	Medium	Admin	Modified

ID	Description	Priority	Stakeholder	Change
FR-31	The system shall allow the administrator to view and manage all registered customer accounts.	Medium	Admin	Retained
FR-47	The system shall provide a notification centre with an audio chime, unread badge counter, and notifications targeted to individual user rooms in real time.	High	Customer / Admin	New
FR-48	The system shall allow the administrator to monitor live station occupancy and slot state without reloading the page.	Medium	Admin	New

Table 11: Functional Requirements – Administrative Dashboard and Notifications

Chapter 4 Non-Functional Requirements

All non-functional requirements are categorised in accordance with ISO/IEC 25010 and include measurable acceptance criteria. NFR IDs from version 02 are preserved; modified criteria are marked, and new requirements continue from NFR-19.

4.1 Performance

ID	Description	Acceptance Criterion	Change
NFR-01	The system shall respond to requests efficiently.	95% of API requests shall respond within 200 ms and the Angular frontend shall reach First Contentful Paint within 1.2 s under normal load; 95% of full page loads shall complete within 3 s.	Modified
NFR-02	The system shall handle concurrent users without degradation.	The system shall support at least 100 concurrent users without a measurable performance drop.	Retained
NFR-03	Slot availability and slot-lock changes shall be reflected in real time.	Slot lock, unlock, and booked events shall be reflected on all connected clients within 500 ms via WebSockets.	Modified

Table 12: Non-Functional Requirements – Performance

4.2 Reliability and Fault Tolerance

ID	Description	Acceptance Criterion	Change
NFR-04	The system shall maintain high availability.	The system shall achieve a minimum uptime of 99% per month, excluding planned maintenance windows.	Retained
NFR-05	The system shall recover from failures gracefully.	In the event of an unexpected failure, the system shall restore full functionality within 30 minutes.	Retained
NFR-19	The system shall prevent failures in one service from cascading to others.	Circuit breakers on inter-service clients shall open after 3 consecutive failures and return fallback responses; no cascading crash shall occur across services.	New
NFR-20	The system shall maintain data consistency across independent service databases.	Failed multi-service operations shall be compensated through Saga compensating transactions so that loyalty balances, slot holds, and payments reach a consistent state (eventual consistency).	New

Table 13: Non-Functional Requirements – Reliability and Fault Tolerance

4.3 Usability

ID	Description	Acceptance Criterion	Change
NFR-06	The system shall be easy to use for first-time users.	A first-time user shall be able to complete a slot booking within 5 minutes without external assistance.	Retained
NFR-07	The system shall support responsive design.	The system shall render correctly on desktop, tablet, and mobile screen sizes (minimum 320 px width).	Retained

ID	Description	Acceptance Criterion	Change
NFR-08	The system shall display error messages clearly.	All user-facing error messages shall be written in plain English and suggest a corrective action; policy errors (for example, cancellation after the cutoff) shall be mapped to specific messages.	Modified
NFR-24	The user interface shall be accessible.	The interface shall conform to WCAG 2.1 Level AA.	New

Table 14: Non-Functional Requirements – Usability

4.4 Security

ID	Description	Acceptance Criterion	Change
NFR-09	All data transmissions shall be encrypted.	The system shall enforce HTTPS (TLS 1.2 or higher) for all communications.	Retained
NFR-10	User passwords shall be stored securely.	Passwords shall be hashed using bcrypt or an equivalent algorithm; plain-text storage is prohibited.	Retained
NFR-11	The system shall enforce role-based access control.	Administrative functions shall be accessible only to authenticated admin accounts; customer accounts shall not access admin panels.	Retained
NFR-12	The system shall protect against common web vulnerabilities.	The system shall pass an OWASP Top 10 vulnerability assessment prior to production deployment.	Retained
NFR-21	The API gateway shall rate-limit authentication endpoints.	A token-bucket limiter on `/api/auth/**` shall allow 5 requests/second per client IP with a burst of 10; excess requests shall receive HTTP 429 with a Retry-After header.	New
NFR-22	Payment data shall be protected.	Card details shall never be stored in Gaming Castle databases; the PayHere IPN endpoint shall accept only notifications with a valid MD5 signature.	New

Table 15: Non-Functional Requirements – Security

4.5 Scalability

ID	Description	Acceptance Criterion	Change
NFR-13	The system shall be designed to accommodate future growth.	The system architecture shall support horizontal scaling to handle a 3x increase in user load without a redesign.	Retained
NFR-14	The database shall handle increasing data volumes efficiently.	Query response times shall remain within 2 seconds under normal conditions as the database grows to 100,000 booking records.	Retained

Table 16: Non-Functional Requirements – Scalability

4.6 Maintainability

ID	Description	Acceptance Criterion	Change
NFR-15	The codebase shall be well-documented.	All modules, functions, and APIs shall include inline comments and a README; documentation coverage shall exceed 80%.	Retained
NFR-16	The system shall support modular architecture.	Each major feature shall be an independently deployable service with its own database (user, booking, payment, tournament, loyalty, notification).	Modified
NFR-23	The system shall be verified by automated tests.	Frontend code coverage shall exceed 80% with 695+ automated unit tests passing on the CI pipeline; backend services shall be verified with unit and integration tests, including Saga failure scenarios.	New

Table 17: Non-Functional Requirements – Maintainability

4.7 Portability

ID	Description	Acceptance Criterion	Change
NFR-17	The system shall operate across major web browsers.	The system shall function correctly on the latest versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.	Retained
NFR-18	The system shall be deployable on standard cloud infrastructure.	The system shall be deployable on platforms such as AWS, Google Cloud, or Azure with minimal configuration changes (Docker containers).	Retained

Table 18: Non-Functional Requirements – Portability

Chapter 5 Business Rules, Constraints, and Assumptions

5.1 Business Rules

- BR-01: A customer must be registered and logged in to make an online slot booking.
- BR-02: Each station may only be assigned to one customer per time slot.
- BR-03: Slot bookings shall be available in minimum increments of one hour, aligned to standard hour boundaries (for example, 10:00–11:00); consecutive hours may be combined in one booking.
- BR-04: Bookings may be cancelled or rescheduled only more than two hours before the slot start time. Eligible cancellations are released immediately and are eligible for a full refund or rescheduling. After the cutoff the system returns the error code `TOO_LATE`.
- BR-05: Loyalty points shall be awarded only upon completion and confirmation of a paid booking.
- BR-06: Tournament registration shall close once the maximum participant limit is reached or the registration deadline passes.
- BR-07: Override capability shall be provided to the administrator to allow any booking to be cancelled or reassigned.
- BR-08: Walk-in customer bookings recorded by the admin shall default to cash payment.
- BR-09 (5-Minute Checkout Hold): Proceeding to checkout shall hold the slot for exactly 300 seconds. If payment is not finalised in that time, the slot shall be released automatically and a `slot_unlocked` event broadcast.
- BR-10 (Loyalty Point Reversal): Loyalty points awarded or redeemed for a booking shall be reversed if that booking is cancelled.
- BR-11 (Game Assignment): A booking shall specify a valid, active game that is compatible with the selected station.
- BR-12 (Rate Limiting): Authentication routes shall accept at most 5 requests per second per IP address with a burst of 10; excess requests receive HTTP 429.
- BR-13 (Review Eligibility): Only the booking owner may review, only completed bookings may be reviewed, and each booking may have at most one review (rating 1–5).
- BR-14 (Reschedule Window): A confirmed booking may be rescheduled to any available slot within 30 days of the request, free of charge, subject to BR-04.
- BR-15 (Station Pricing): The charge for a booking shall equal the number of booked hours multiplied by the hourly rate of the selected station, less any loyalty discount.

5.2 Constraints

- The system must be developed as a web application accessible via standard web browsers.
- The initial deployment shall support PS5, PC Esports, and VR station categories; further stations of any category can be added by the administrator without code changes.
- The online payment gateway shall be PayHere, and all online transactions shall be in LKR.
- The project must be completed within the academic semester timeline.
- The development team consists of 4 members; resource allocation is constrained accordingly.
- The system shall be built using open-source or freely available frameworks to minimise licensing costs.
- Internet connectivity is required for the system to function; offline mode is not within the project scope.
- Real-time slot holds are maintained in the memory of the real-time server; the database unique constraint remains the authoritative guard against double-booking.

5.3 Assumptions

- It is assumed that Gaming Castle management will provide timely feedback and sign-off at each project milestone.
- It is assumed that all customers have access to a device with internet connectivity and a modern web browser.
- It is assumed that the hourly rates and game titles will be provided by the client.
- It is assumed that a PayHere merchant account is available and that the PayHere IPN callback can reach the system's public endpoint.
- It is assumed that SMTP and SMS (Twilio) providers are available for notifications and verification codes.
- It is assumed that the Gaming Castle premises will have stable internet connectivity for admin operations.
- It is assumed that sufficient predefined rules or sample data will be available to configure or simulate the automated tournament management module.

Chapter 6 Use Case Diagrams and Descriptions

6.1 System Actors

The Gaming Castle system involves two primary actors and two secondary actors:

- Customer (Registered User): A person who accesses the system online to book and manage slots, hold slots during checkout, pay via PayHere, register for tournaments, redeem loyalty rewards, and review completed sessions.
- Administrator (Admin): A member of Gaming Castle staff who manages bookings (including walk-ins), stations and games, tournaments, payments, and customer accounts through the admin dashboard.
- System / Gateway (secondary actor): Enforces rate limiting and token validation, broadcasts real-time events, receives PayHere IPN callbacks, and executes Saga compensations.
- PayHere (external actor): Processes payments and sends IPN notifications.

6.2 Use Case Summary

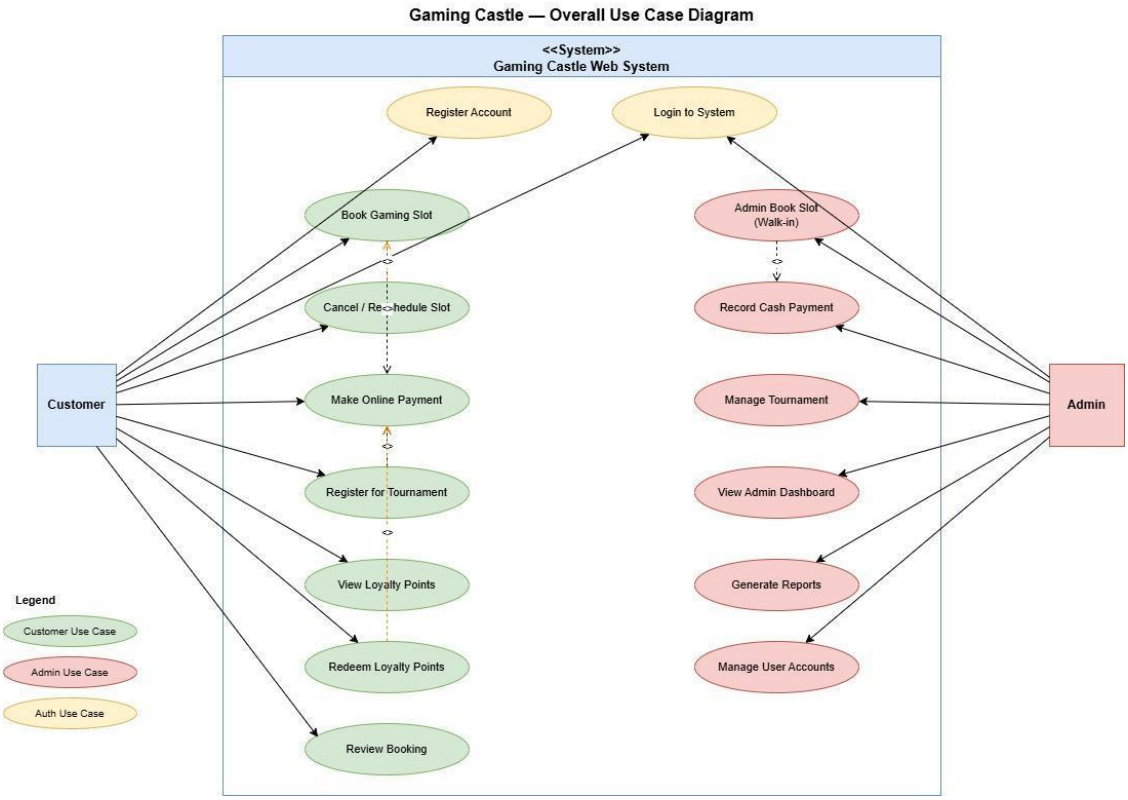


Figure 1: Overall Use Case Diagram – Gaming Castle Web System

Note: diagram shown is from version 02 and is pending update to match version 03.

Use Case ID	Use Case Name	Actor(s)	Priority	Change
UC-01	Register Account	Customer	High	Retained
UC-02	Login to the System	Customer / Admin	High	Modified

Use Case ID	Use Case Name	Actor(s)	Priority	Change
UC-03	Book Gaming Slot	Customer	High	Modified
UC-04	Cancel / Reschedule Slot	Customer	High	Modified
UC-05	Admin Book Slot for Walk-in Customer	Admin	High	Modified
UC-06	Make Online Payment	Customer, PayHere	High	Modified
UC-07	Record Cash Payment	Admin	High	Retained
UC-08	Register for Tournament	Customer	High	Modified
UC-09	Manage Tournament	Admin	High	Modified
UC-10	View Loyalty Points	Customer	Medium	Modified
UC-11	Redeem Loyalty Points	Customer	Medium	Modified
UC-12	View Admin Dashboard	Admin	High	Modified
UC-13	Generate Reports	Admin	Medium	Retained
UC-14	Manage User Accounts	Admin	Medium	Modified
UC-15	Submit Session Review	Customer	Medium	New
UC-16	Manage Stations and Games	Admin	Medium	New

Table 19: Use Case Summary

6.3 Use Case Descriptions

Use Case UC-01: Register Account

Attribute	Details
Use Case ID	UC-01
Use Case Name	Register Account
Actor(s)	Customer
Preconditions	The customer has not previously registered. The system is accessible.
Main Flow	1. The customer navigates to the Register page. 2. Required details are entered (name, email, phone number, password). 3. The input fields are validated by the system. 4. It is verified by the system that the email or phone number is not already registered. 5. The password is hashed (bcrypt), the account is created with role CUSTOMER, and a loyalty account with 0 points is initialised. 6. A confirmation email is sent and the user is redirected to the Login page.
Alternative Flow	3a. If any field is invalid – the error is highlighted and correction is prompted by the system. 4a. If the email or phone number is already registered – the following message is displayed: An account with this email already exists.
Postconditions	A new customer account is created and stored in the system. The customer may now log in.

Attribute	Details
Exception Flow	If the system is unavailable, a service error message is displayed and the customer is prompted to try again later. If the rate limit is exceeded, the user is asked to wait and retry (HTTP 429).
Related Requirements	FR-01, FR-02, FR-03, NFR-21

Table 20: Use Case Description – UC-01: Register Account

Use Case UC-02: Login to System

Attribute	Details
Use Case ID	UC-02
Use Case Name	Login to System
Actor(s)	Customer / Admin
Preconditions	The user has a registered, active account. The system is accessible.
Main Flow	1. The user navigates to the Login page. 2. An email address or phone number and password are entered by the user. 3. The gateway applies rate limiting and the credentials are validated by the user service. 4. The user is authenticated and the role (Customer or Admin) is identified by the system. 5. The user is redirected to the appropriate dashboard (Customer home or Admin dashboard).
Alternative Flow	3a. If credentials are invalid – an error message is displayed, the failed-attempt counter is incremented, and after 5 consecutive failures the account is temporarily locked. 3b. If the password has been forgotten – the user selects Forgot Password and is redirected to the password reset flow using an email or SMS verification code.
Postconditions	The user is authenticated and granted access to role-appropriate features.
Exception Flow	If the account is deactivated, the following message is displayed: Your account has been deactivated, with an option to start reactivation. If the rate limit is exceeded, HTTP 429 is returned.
Related Requirements	FR-02, FR-03, FR-04, FR-05, FR-32, NFR-21

Table 21: Use Case Description – UC-02: Login to System

Use Case UC-03: Book Gaming Slot

Attribute	Details
Use Case ID	UC-03
Use Case Name	Book Gaming Slot
Actor(s)	Customer
Preconditions	The customer is registered and logged into the system.

Attribute	Details
Main Flow	1. The customer navigates to the Book a Slot page. 2. A date and a station category (PS5, PC, VR) are selected; station cards show average ratings. 3. A station, one or more consecutive hourly slots, and a compatible game are selected. 4. The booking summary with the applicable charge (hours × station rate) is displayed by the system. 5. The customer proceeds to checkout; the slot is soft-locked for 5 minutes and all clients see "Holding". 6. Loyalty points may optionally be applied (UC-11) and payment is made (UC-06). 7. On payment confirmation the booking becomes CONFIRMED, the slot is shown as booked on all clients, and a confirmation is sent by email and SMS.
Alternative Flow	3a. If the slot is unavailable or held by another customer – the slot is disabled and alternative times are suggested. 5a. If the hold expires – the slot is released and the customer is asked to reselect.
Postconditions	A confirmed booking record is created. The slot is marked as unavailable for other users.
Exception Flow	If payment fails, the slot hold is released, any loyalty redemption is compensated, and the customer is notified.
Related Requirements	FR-07, FR-08, FR-10, FR-12, FR-19, FR-20, FR-21, FR-34, NFR-03

Table 22: Use Case Description – UC-03: Book Gaming Slot

Use Case UC-04: Cancel / Reschedule Slot

Attribute	Details
Use Case ID	UC-04
Use Case Name	Cancel / Reschedule Slot
Actor(s)	Customer
Preconditions	The customer is registered and logged in. A CONFIRMED booking is held by the customer.
Main Flow	1. The customer navigates to My Bookings. 2. A booking to cancel or reschedule is selected. 3. The system checks that the start time is more than 2 hours away. 4. To cancel: the customer confirms, the booking becomes CANCELLED, the slot is released immediately, loyalty points are reversed, and a refund is initiated where applicable. 5. To reschedule: the customer selects any available slot within 30 days and the booking is moved at no fee. 6. The customer is notified of the change via email or SMS.
Alternative Flow	5a. If rescheduling – available slots are displayed for the customer to choose from.
Postconditions	The original slot is released and made available for other users. The booking record is updated or cancelled and the loyalty balance reflects any reversal.
Exception Flow	If the start time is less than 2 hours away, the request is refused with error code TOO_LATE and the customer is informed that the booking can no longer be modified.
Related Requirements	FR-11, FR-12, FR-35, FR-45, FR-46

Table 23: Use Case Description – UC-04: Cancel / Reschedule Slot

Use Case UC-05: Admin Book Slot for Walk-in Customer

Attribute	Details
Use Case ID	UC-05
Use Case Name	Admin Book Slot for Walk-in Customer
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access. At least one slot is available.
Main Flow	1. The admin navigates to the Admin Booking panel. 2. An available date, time, station, and game are selected. 3. The walk-in customer details are entered by the admin. 4. A WALKIN booking is created and the slot is marked as occupied by the system. 5. The cash payment is recorded by the admin (UC-07) and the booking is confirmed immediately. 6. Revenue records are updated and a printable receipt is generated by the system.
Alternative Flow	2a. If no slots are available for the selected time – the admin is notified and alternative times are suggested.
Postconditions	A booking record is created for the walk-in customer. The slot is marked as occupied. Cash payment is recorded.
Exception Flow	If the system fails to save the booking, an error message is displayed and the admin is prompted to retry.
Related Requirements	FR-09, FR-21, FR-22

Table 24: Use Case Description – UC-05: Admin Book Slot for Walk-in Customer

Use Case UC-06: Make Online Payment

Attribute	Details
Use Case ID	UC-06
Use Case Name	Make Online Payment
Actor(s)	Customer, PayHere
Preconditions	The customer is registered and logged in. A slot is held for a pending booking.
Main Flow	1. The booking summary with the total charge in LKR is displayed by the system. 2. Card details are entered; the form formats input, detects the brand, and validates the number with the Luhn check. 3. A payment token is requested and the PayHere MD5 signature is computed by the payment service. 4. The customer submits payment to PayHere. 5. PayHere sends an IPN webhook and the system validates its signature. 6. The payment is marked COMPLETED, the booking is confirmed, loyalty points are awarded, and a digital receipt is displayed. 7. A booking confirmation is sent by email and SMS.
Alternative Flow	4a. If the customer abandons the payment – the hold expires after 5 minutes and the slot is released.

Attribute	Details
Postconditions	Payment is confirmed. The booking is finalised. Loyalty points are awarded to the customer.
Exception Flow	If payment fails, or a downstream step fails after capture, Saga compensating transactions (release hold, reverse loyalty, refund) are run and the customer is notified. If a dependent service is unavailable, its circuit breaker returns a fallback and the operation is retried or compensated.
Related Requirements	FR-19, FR-20, FR-21, FR-23, FR-24, FR-42, FR-43, FR-44, NFR-19, NFR-20, NFR-22

Table 25: Use Case Description – UC-06: Make Online Payment

Use Case UC-07: Record Cash Payment

Attribute	Details
Use Case ID	UC-07
Use Case Name	Record Cash Payment
Actor(s)	Admin
Preconditions	The admin is authenticated. A walk-in booking has been created by the admin.
Main Flow	1. The admin navigates to the payment recording section. 2. The relevant booking is selected. 3. The cash amount received is entered by the admin. 4. The amount is validated by the system against the booking charge. 5. The booking is marked as paid by the system. 6. A receipt is generated and revenue records are updated.
Alternative Flow	3a. If the amount entered does not match the expected charge – the admin is alerted to confirm before proceeding.
Postconditions	The booking is marked as paid (cash). Revenue records are updated. A receipt is generated.
Exception Flow	If the system fails to record the payment, an error is displayed and the admin is prompted to retry.
Related Requirements	FR-21, FR-22

Table 26: Use Case Description – UC-07: Record Cash Payment

Use Case UC-08: Register for Tournament

Attribute	Details
Use Case ID	UC-08
Use Case Name	Register for Tournament
Actor(s)	Customer
Preconditions	The customer is registered and logged in. A tournament with open registration exists.

Attribute	Details
Main Flow	1. The customer navigates to the Tournaments page; the ticker banner shows status, prize pool, and spots remaining. 2. A tournament is selected and its details are viewed (game, date, entry fee, max participants). 3. The customer selects Register and pays the entry fee (UC-06). 4. Registration is confirmed and the participant count is incremented. 5. A confirmation notification is sent to the customer.
Alternative Flow	3a. If payment fails – registration is not confirmed and the customer is notified.
Postconditions	The customer is registered for the tournament. Entry fee payment is recorded.
Exception Flow	If the tournament is full or the registration deadline has passed – the customer is informed that registration is closed.
Related Requirements	FR-14, FR-19, FR-20, FR-40

Table 27: Use Case Description – UC-08: Register for Tournament

Use Case UC-09: Manage Tournament

Attribute	Details
Use Case ID	UC-09
Use Case Name	Manage Tournament
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the Tournament Management section. 2. A tournament is created or edited by entering details (name, game, date, entry fee, max participants, cover image). 3. The tournament is published, made visible to customers, and shown in the ticker banner by the system. 4. Registrations are monitored in real-time by the admin. 5. Tournament status is updated as it progresses. 6. Tournament results are recorded and published.
Alternative Flow	The admin may cancel a tournament before it begins; all registered participants are notified automatically by email and by a real-time browser alert.
Postconditions	Tournament records are created or updated in the system.
Exception Flow	If the maximum participant limit is reached, registration is automatically closed by the system.
Related Requirements	FR-13, FR-15, FR-16, FR-17, FR-18, FR-40, FR-41

Table 28: Use Case Description – UC-09: Manage Tournament

Use Case UC-10: View Loyalty Points

Attribute	Details
Use Case ID	UC-10
Use Case Name	View Loyalty Points
Actor(s)	Customer

Attribute	Details
Preconditions	The customer is registered and logged in.
Main Flow	1. The customer navigates to their Profile or Loyalty Rewards section. 2. The loyalty balance and tier are retrieved by the system. 3. The current balance, tier, and a history of EARNED, REDEEMED, and REVERSED transactions with running balance are displayed.
Alternative Flow	3a. If the customer has no points history – a balance of 0 is displayed along with a message encouraging the first booking.
Postconditions	The current loyalty points balance and transaction history are viewed by the customer.
Exception Flow	If the loyalty data cannot be loaded (for example, the loyalty service is unavailable), a fallback message is displayed and the customer is prompted to refresh the page.
Related Requirements	FR-26, FR-46

Table 29: Use Case Description – UC-10: View Loyalty Points

Use Case UC-11: Redeem Loyalty Points

Attribute	Details
Use Case ID	UC-11
Use Case Name	Redeem Loyalty Points
Actor(s)	Customer
Preconditions	The customer is registered and logged in. Sufficient loyalty points are available.
Main Flow	1. The customer proceeds to checkout for a held slot. 2. The available loyalty points and equivalent discount are displayed by the system. 3. The customer opts to redeem points against the booking charge. 4. The discount is applied and the updated total is displayed. 5. Payment for the remaining balance is confirmed and completed. 6. The redeemed points are deducted from the customer's balance and a REDEEMED transaction is logged.
Alternative Flow	3a. If redemption is not opted for – the booking proceeds at full price with no points deducted.
Postconditions	Loyalty points are deducted. The booking charge is reduced by the redeemed amount.
Exception Flow	If payment fails or the booking is later cancelled, the redeemed points are restored through a compensating transaction. If the balance drops below the minimum redemption threshold during checkout, the option is disabled and the customer is notified.
Related Requirements	FR-24, FR-25, FR-45, FR-46

Table 30: Use Case Description – UC-11: Redeem Loyalty Points

Use Case UC-12: View Admin Dashboard

Attribute	Details
Use Case ID	UC-12
Use Case Name	View Admin Dashboard

Attribute	Details
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin logs in and is redirected to the Admin Dashboard. 2. Real-time summary widgets are displayed: active sessions, today's revenue, station utilisation, upcoming bookings, and the tournament ticker. 3. The notification centre shows unread notifications with an audio chime for new events. 4. The admin navigates to detailed sections (Bookings, Payments, Tournaments, Users, Stations and Games, Reports) from the dashboard.
Alternative Flow	2a. If there are no active sessions or bookings – empty state messages are displayed for the relevant widgets.
Postconditions	A real-time operational overview of the Gaming Castle system is viewed by the admin.
Exception Flow	If real-time data fails to load, the last cached data is displayed with a warning that data may not be current.
Related Requirements	FR-28, FR-40, FR-47, FR-48

Table 31: Use Case Description – UC-12: View Admin Dashboard

Use Case UC-13: Generate Reports

Attribute	Details
Use Case ID	UC-13
Use Case Name	Generate Reports
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the Reports section from the Admin Dashboard. 2. A report type is selected (booking revenue, station utilisation, tournament participation). 3. The date range and any filters are specified. 4. Relevant data is retrieved and processed by the system. 5. The report is displayed in a structured format. 6. The report may be exported (CSV or PDF) by the admin.
Alternative Flow	5a. If no data is available for the selected criteria – the admin is informed that no records match the filter.
Postconditions	A report is generated and optionally exported by the admin.
Exception Flow	If report generation fails due to a system error, the admin is notified and prompted to retry.
Related Requirements	FR-29

Table 32: Use Case Description – UC-13: Generate Reports

Use Case UC-14: Manage User Accounts

Attribute	Details
Use Case ID	UC-14

Attribute	Details
Use Case Name	Manage User Accounts
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin navigates to the User Management section. 2. A list of all registered customer accounts is displayed. 3. Users may be searched or filtered by name, email, or status. 4. A user is selected to view their profile, booking history, loyalty points, and reviews. 5. Account details may be updated or the account may be activated or deactivated if required. 6. Changes are saved and reflected immediately by the system.
Alternative Flow	5a. If a user is deactivated – a notification is sent to the affected user by the system.
Postconditions	User account details are updated or the account is activated or deactivated as required.
Exception Flow	If the system fails to save changes, an error is displayed and the admin is prompted to retry.
Related Requirements	FR-06, FR-31, FR-39

Table 33: Use Case Description – UC-14: Manage User Accounts

Use Case UC-15: Submit Session Review

Attribute	Details
Use Case ID	UC-15
Use Case Name	Submit Session Review
Actor(s)	Customer
Preconditions	The customer is logged in and owns a COMPLETED booking that has no existing review.
Main Flow	1. The customer opens My Bookings, selects a completed booking, and chooses Rate Session. 2. The five-star review modal is opened by the system. 3. A rating from 1 to 5 is selected and optional feedback is entered (for example, about the DualSense controller or the rig). 4. Eligibility is validated, the review is saved, and the station's average rating is updated by the system. 5. The review appears in My Reviews and in the administrator's review view.
Alternative Flow	The customer closes the modal without saving; nothing is stored.
Postconditions	One review is linked to the booking and station ratings are updated.
Exception Flow	If the booking is cancelled, upcoming, not owned by the customer, or already reviewed, the submission is rejected with an explanatory message.
Related Requirements	FR-36, FR-37, FR-38, FR-39, BR-13

Table 34: Use Case Description – UC-15: Submit Session Review

Use Case UC-16: Manage Stations and Games

Attribute	Details
Use Case ID	UC-16
Use Case Name	Manage Stations and Games
Actor(s)	Admin
Preconditions	The admin is authenticated and has admin-level access.
Main Flow	1. The admin opens Station and Game Management. 2. A station (code, name, type CONSOLE/PC/VR, hourly rate, image) or a game (title, genre, cover, platforms, featured flag) is added or edited. 3. Games are assigned to stations and stations and games are activated or deactivated. 4. The changes are saved and customer availability and pricing views are updated.
Alternative Flow	Deactivating a station with future confirmed bookings prompts the admin to reassign or cancel those bookings (BR-07).
Postconditions	The station and game catalogue is updated.
Exception Flow	Duplicate station codes or invalid rates are rejected with an error message.
Related Requirements	FR-07, FR-19, FR-30, BR-11, BR-15

Table 35: Use Case Description – UC-16: Manage Stations and Games

Chapter 7 Activity Diagrams

This chapter presents ten key workflows within the Gaming Castle system. Each activity diagram illustrates the sequence of activities, decision points, and possible outcomes for a primary use case.

7.1 Slot Booking

This activity diagram describes the process by which a registered customer books a gaming slot on a PS5, PC, or VR station through the Gaming Castle web system. The workflow covers slot availability checking, the real-time slot hold, payment processing, and booking confirmation.

Step	Activity	Actor	Decision / Outcome
1	Customer Logs In	Customer	Precondition: Customer must be registered
2	Navigate to Book a Slot Page; select date and station category (PS5 / PC / VR)	Customer	–
3	Select station, consecutive hourly slots, and game	Customer	–
4	Check Slot Availability	System	Decision: Slot Available and not held? YES → Step 5 | NO → Show message, suggest alternatives → END
5	View Booking Summary and Charge (hours × station rate)	System	–
6	Proceed to checkout; acquire 5-minute soft lock	System	Broadcast slot_locked; other clients show "Holding"
7	Optionally apply loyalty points	Customer	See 7.4
8	Pay online (PayHere)	Customer	See 7.7
9	Payment outcome	System	Decision: Success → Step 10 | Failure or hold expired → Release lock, broadcast slot_unlocked, compensate, notify customer → END
10	Booking Confirmed	System	Broadcast slot_booked_live
11	Send Confirmation Notification (email / SMS)	System	Postcondition: Slot marked unavailable

Table 36: Activity Steps – Slot Booking

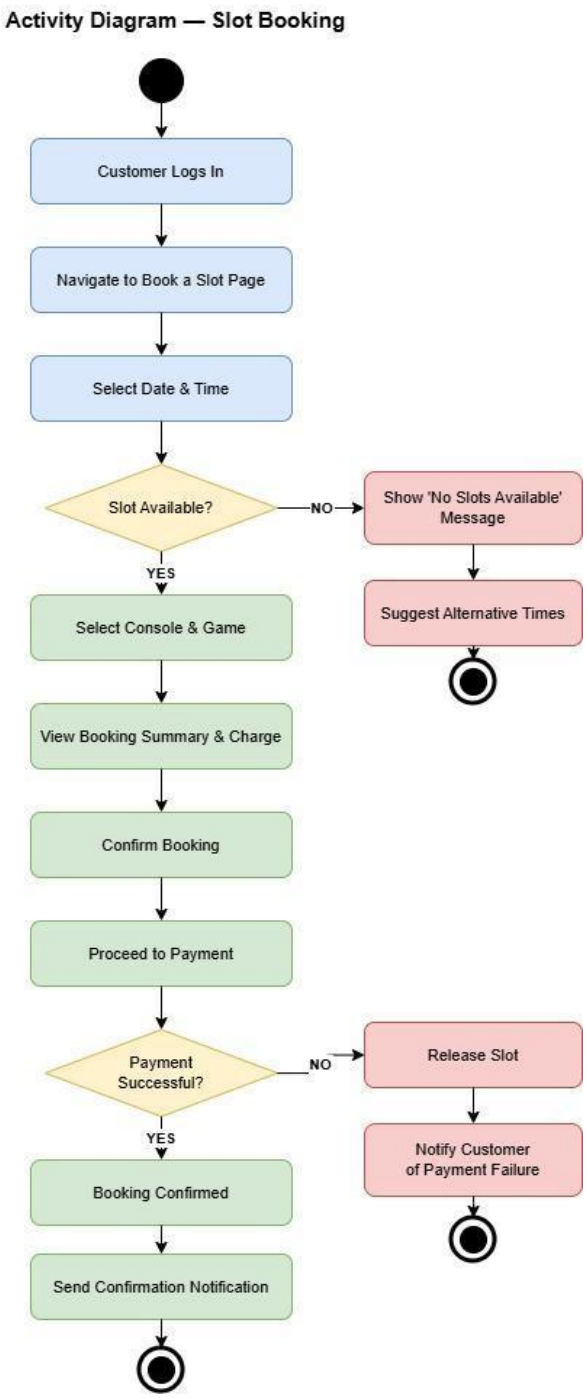


Figure 2: Activity Diagram – Slot Booking

Note: diagram shown is from version 02 and is pending update to match version 03.

7.2 Tournament Registration

This activity diagram describes the process by which a registered customer views, selects, and registers for a tournament on the Gaming Castle platform, including entry fee payment validation.

Step	Activity	Actor	Decision / Outcome
1	Customer Logs In	Customer	Precondition: Customer must be registered
2	Navigate to Tournaments Page; view list and ticker banner	Customer	–
3	Select a Tournament	Customer	–
4	Check Registration Status	System	Decision: Registration Open? NO → Show Closed message → END
5	Check Participant Limit	System	Decision: Tournament Full? YES → Show Full message → END
6	Click Register; proceed to Entry Fee Payment	Customer	–
7	Process Payment	System	Decision: Payment Successful? YES → Step 8 | NO → Notify customer → END
8	Registration Confirmed; participant count incremented	System	–
9	Send Confirmation Notification	System	Postcondition: Customer on participant list

Table 37: Activity Steps – Tournament Registration

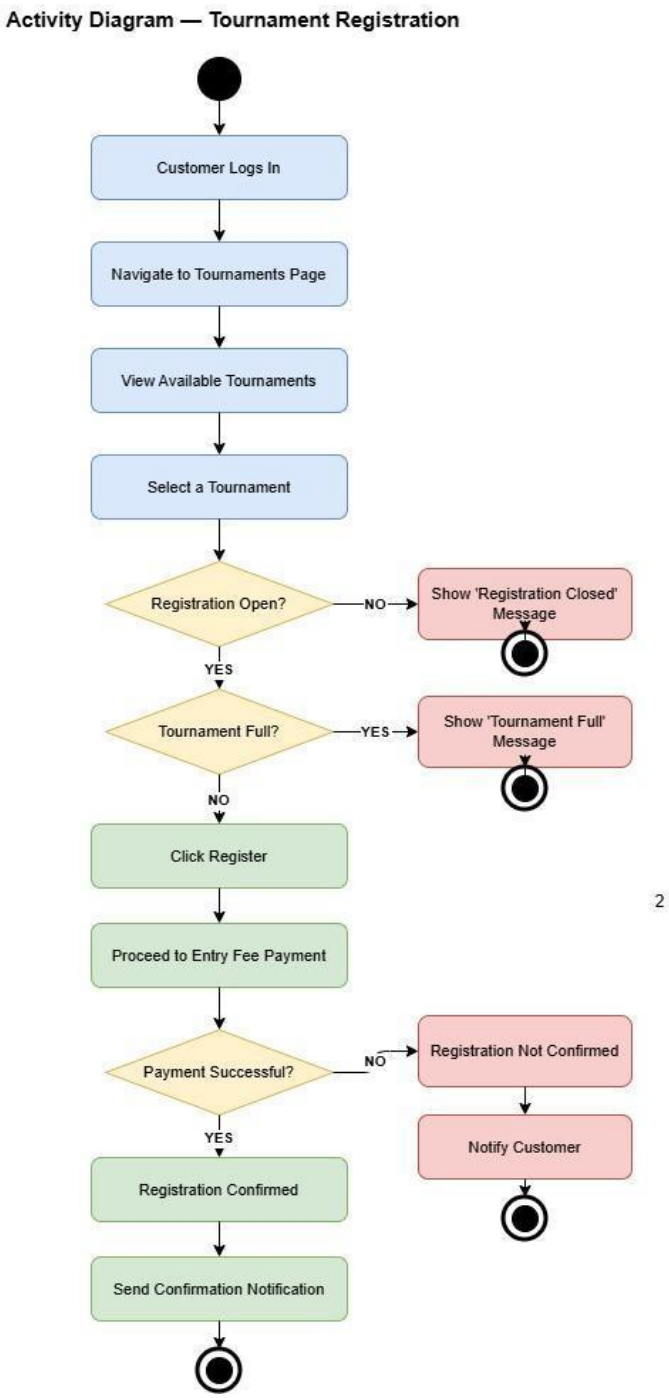


Figure 3: Activity Diagram – Tournament Registration

7.3 Admin Walk-in Booking

This activity diagram describes the process by which an administrator books a gaming slot on behalf of a walk-in customer who does not use the online system, and records a cash payment.

Step	Activity	Actor	Decision / Outcome
1	Admin Logs In	Admin	Precondition: Admin must be authenticated
2	Navigate to Admin Booking Panel; View Available Slots	Admin	–

Step	Activity	Actor	Decision / Outcome
3	Check Slot Availability	System	Decision: Slot Available? NO → Notify admin → END
4	Select Slot, Station and Game; Enter Walk-in Customer Name	Admin	–
5	System Creates WALKIN Booking and marks slot Occupied	System	–
6	Admin Records Cash Payment	Admin	–
7	Confirm booking, update revenue records, generate printable receipt	System	Postcondition: Revenue records updated

Table 38: Activity Steps – Admin Walk-in Booking

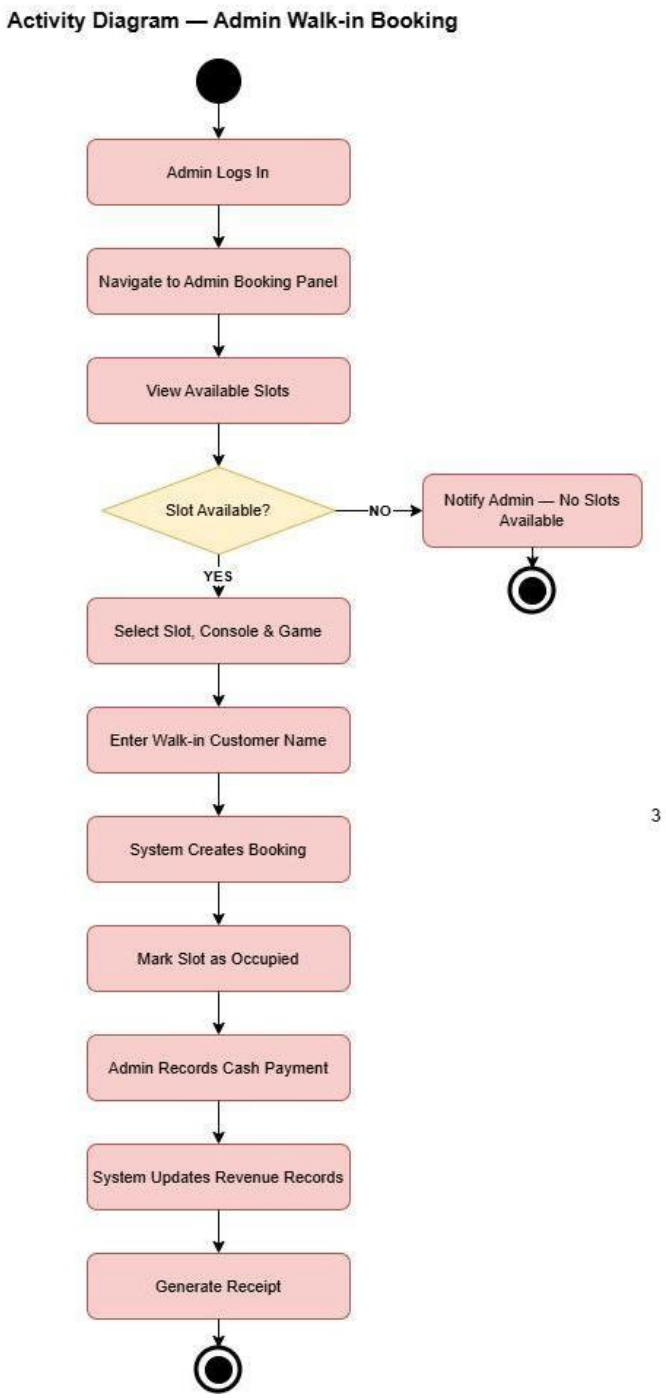


Figure 4: Activity Diagram – Admin Walk-in Booking

7.4 Loyalty Points Redemption

This activity diagram describes the process by which a registered customer applies accumulated loyalty points as a discount during a slot booking payment, including restoration of points on failure.

Step	Activity	Actor	Decision / Outcome
1	Customer proceeds to checkout	Customer	Slot is held
2	View Loyalty Points Balance	System	–

Step	Activity	Actor	Decision / Outcome
3	Check Sufficient Points	System	Decision: Sufficient Points? NO → Proceed to full payment
4	Apply Loyalty Points Discount; View Discounted Booking Summary	System / Customer	–
5	Confirm and Pay Remaining Balance	Customer	–
6	Process Payment	System	Decision: Payment Successful? YES → Step 7 | NO → Restore points via compensating transaction, notify → END
7	Deduct Redeemed Points; log REDEEMED transaction	System	–
8	Booking Confirmed	System	Postcondition: Points balance updated

Table 39: Activity Steps – Loyalty Points Redemption

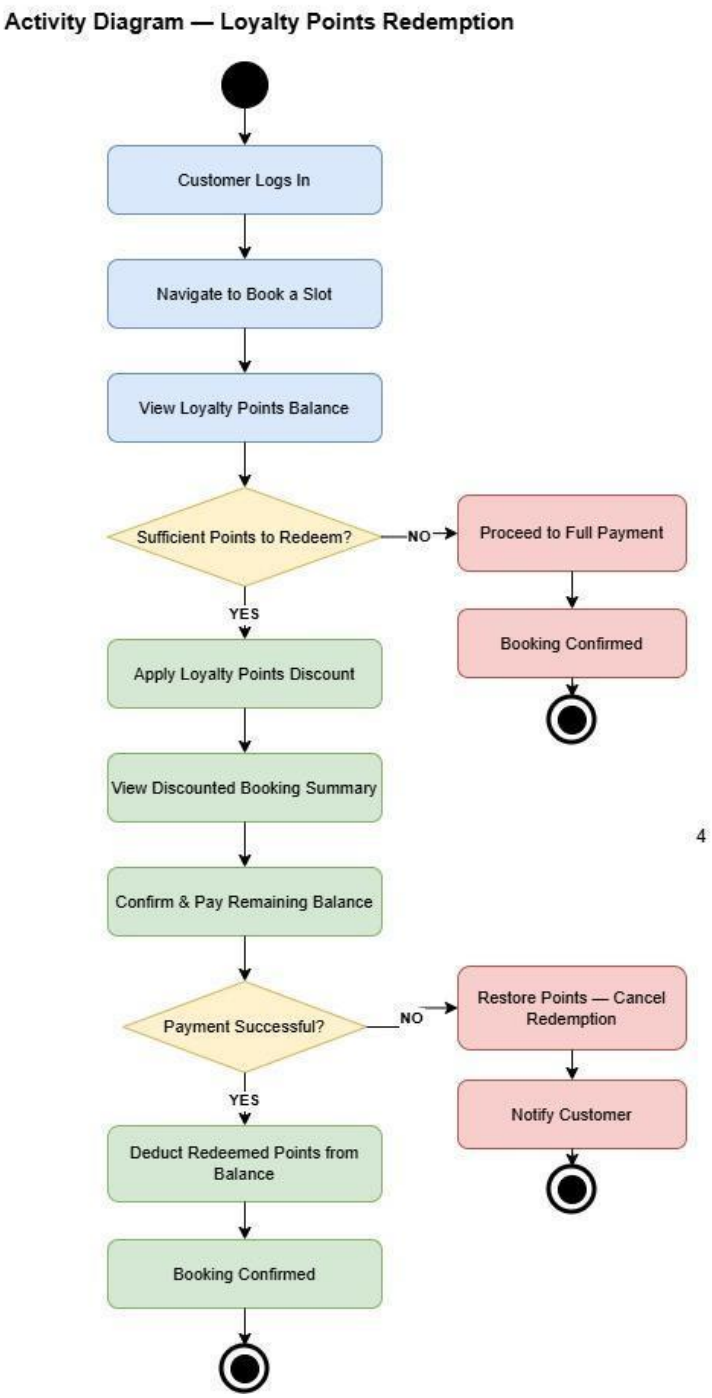


Figure 5: Activity Diagram – Loyalty Points Redemption

Note: diagram shown is from version 02 and is pending update to match version 03.

7.5 User Login / Authentication

This activity diagram describes the authentication process for both customers and administrators accessing the Gaming Castle web system, including gateway rate limiting, credential validation, role-based redirection, and account lockout after repeated failed attempts.

Step	Activity	Actor	Decision / Outcome
1	Open Login Page; enter email or phone and password; submit	User	–
2	Apply Gateway Rate Limit	System	Decision: Within 5 req/s (burst 10)? NO → HTTP 429 → END
3	Validate Credentials	System	Decision: Credentials Valid? YES → Step 5 | NO → Increment failed attempts
4	Check Failed Attempts	System	Decision: Attempts < 5? YES → Retry | NO → Lock account until lockout expiry → END
5	Check Account Active	System	Decision: Deactivated? YES → Offer reactivation → END
6	Verify User Role and Issue Tokens	System	–
7	Role-based Redirect	System	Decision: Role? Customer → Customer Dashboard | Admin → Admin Dashboard

Table 40: Activity Steps – User Login / Authentication

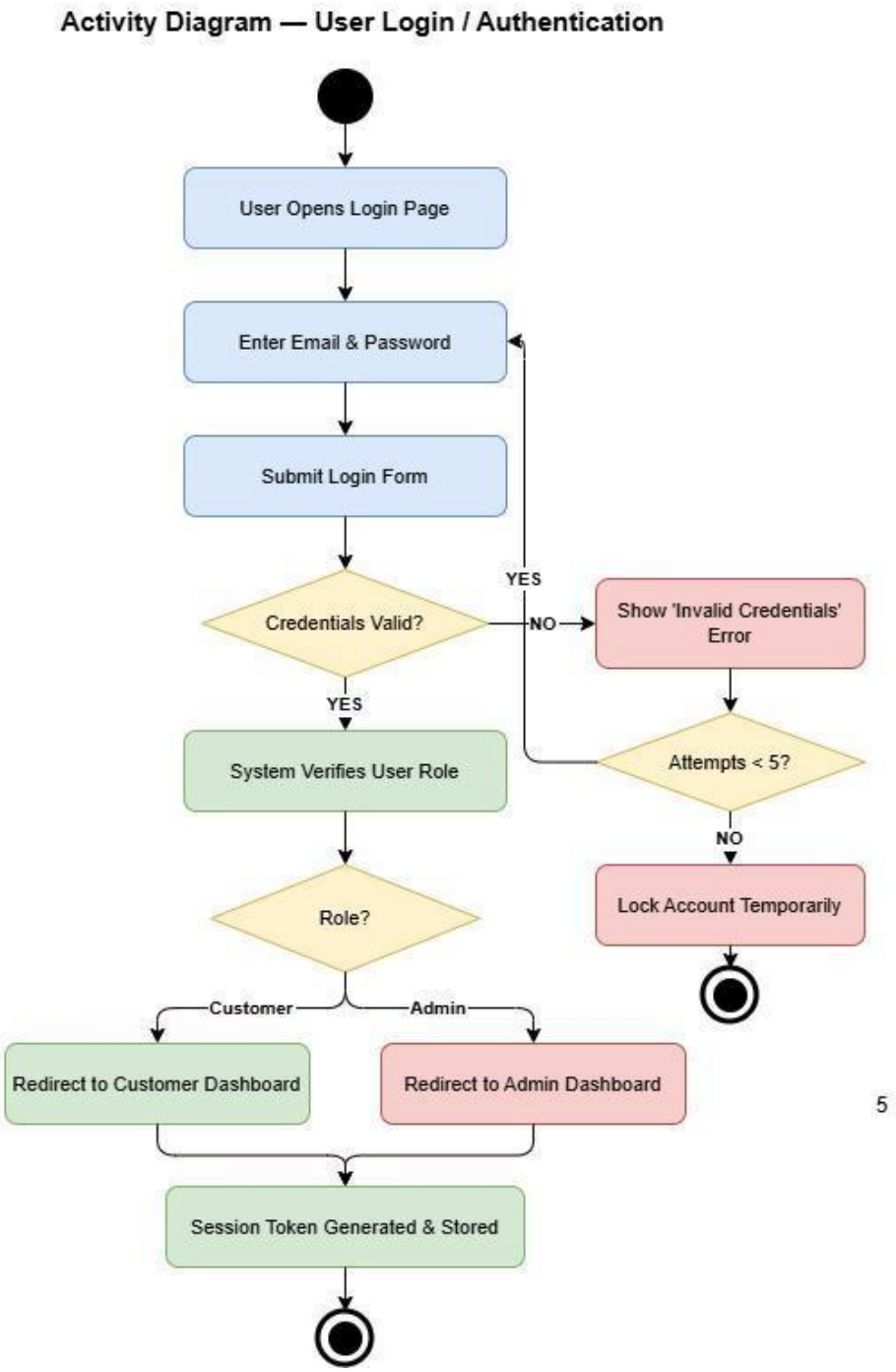


Figure 6: Activity Diagram – User Login / Authentication

Note: diagram shown is from version 02 and is pending update to match version 03.

7.6 User Registration

This activity diagram describes the process by which a new user creates an account on the Gaming Castle web system, including form validation, duplicate checking, account creation, and confirmation email dispatch.

Step	Activity	Actor	Decision / Outcome
1	Open Registration Page; enter Name, Email, Phone and Password; submit	User	–
2	Validate All Fields	System	Decision: All Fields Valid? NO → Show inline errors → END
3	Check Email and Phone Uniqueness	System	Decision: Already Registered? YES → Show error → END
4	Hash Password (bcrypt) and Store Account with role CUSTOMER	System	–
5	Initialise Loyalty Account to 0 Points	System	–
6	Send Confirmation Email; Redirect to Login Page	System	Postcondition: New customer account is active

Table 41: Activity Steps – User Registration

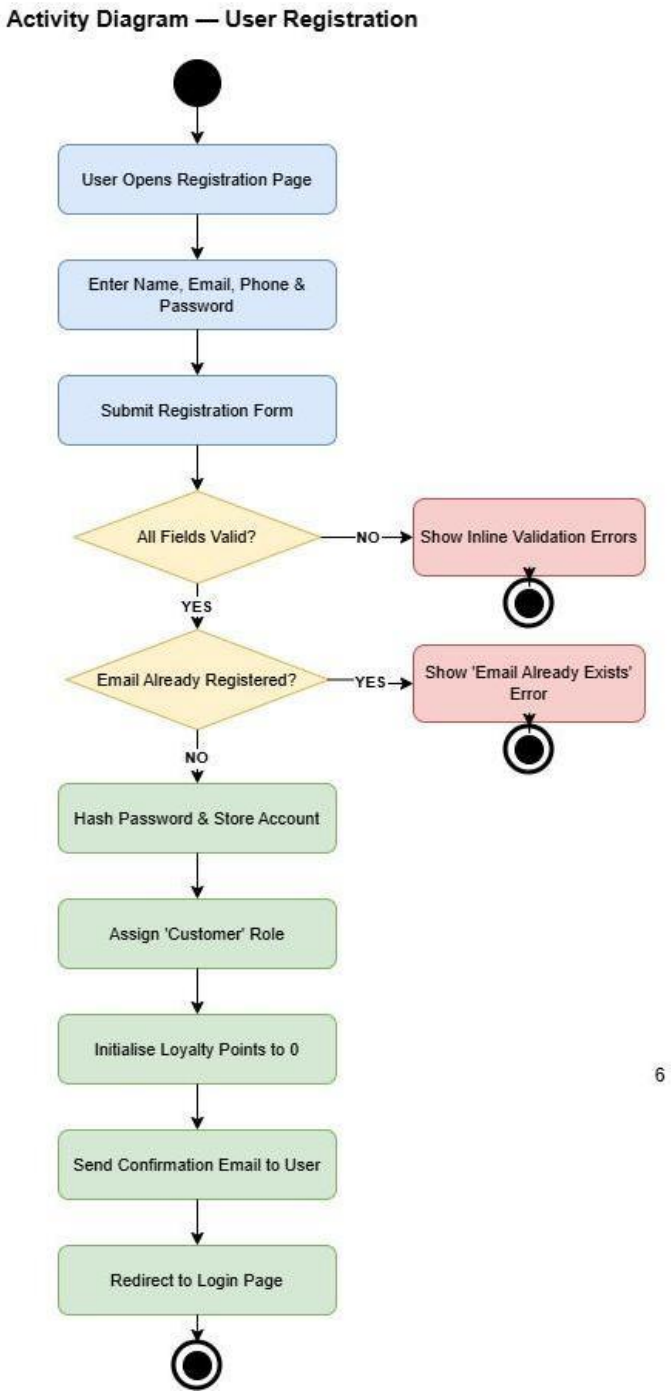


Figure 7: Activity Diagram – User Registration

7.7 Payment Processing

This activity diagram describes the end-to-end payment processing workflow, covering card entry and validation, PayHere payment, asynchronous IPN verification, receipt generation, loyalty point award, and Saga compensation on failure.

Step	Activity	Actor	Decision / Outcome
1	Display Payment Summary (LKR)	System	–

Step	Activity	Actor	Decision / Outcome
2	Enter card details; auto-format, detect brand, Luhn validation	Customer	Decision: Luhn invalid? YES → Show error, block submit
3	Request payment token; compute PayHere MD5 signature	System	–
4	Submit payment to PayHere	Customer	–
5	Await IPN webhook	System	Decision: Hold expired before IPN? YES → Release slot, mark PENDING for reconciliation
6	Validate IPN signature	System	Decision: Valid? NO → Reject and log → END
7	Check payment status	System	Decision: COMPLETED → Step 8 | FAILED → Step 10
8	Confirm booking; record payment; generate and display receipt	System	–
9	Award loyalty points; send email/SMS; broadcast slot_booked_live	System	Postcondition: Payment confirmed, loyalty points updated
10	Run compensating transactions (release hold, restore redeemed points, refund if captured)	System	Record in compensating_transactions; notify customer → END

Table 42: Activity Steps – Payment Processing

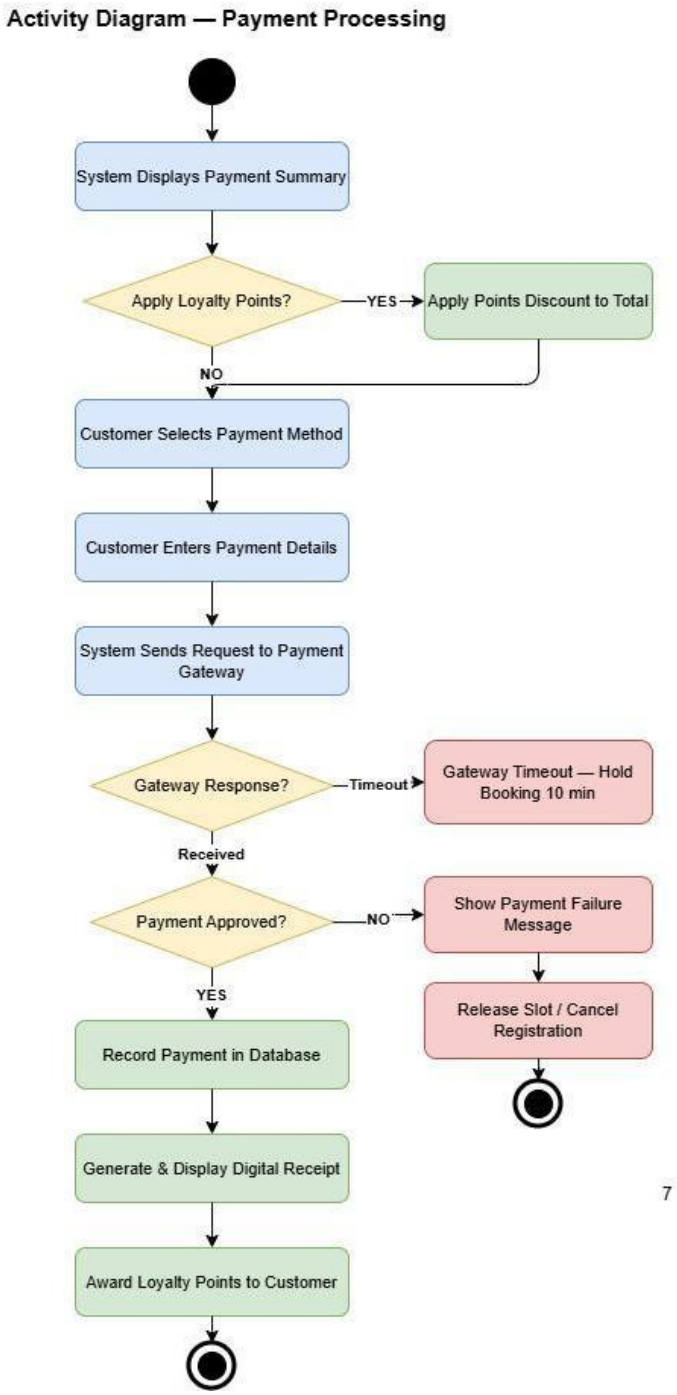


Figure 8: Activity Diagram – Payment Processing

Note: diagram shown is from version 02 and is pending update to match version 03.

7.8 Admin Dashboard Overview

This activity diagram describes the workflow of an administrator interacting with the Gaming Castle admin dashboard, including navigation across key management modules: bookings, tournaments, users, stations and games, and reports.

Step	Activity	Actor	Decision / Outcome
1	Admin Logs In Successfully	Admin	Precondition: Admin authenticated

Step	Activity	Actor	Decision / Outcome
2	System Loads Admin Dashboard	System	Displays: Real-time bookings, revenue summary, active sessions, utilisation, ticker
3	Admin Selects Action	Admin	Decision: Manage Bookings | Manage Tournaments | Manage Users | Stations and Games | Generate Reports
4	Perform action	Admin	View / cancel / create bookings; create / edit / cancel tournaments; manage accounts; manage stations; generate reports
5	Action Completed; notification broadcast if applicable	System	All branches merge here
6	Continue on Dashboard?	Admin	Decision: YES → Refresh dashboard → Step 3 | NO → Logout → END

Table 43: Activity Steps – Admin Dashboard Overview

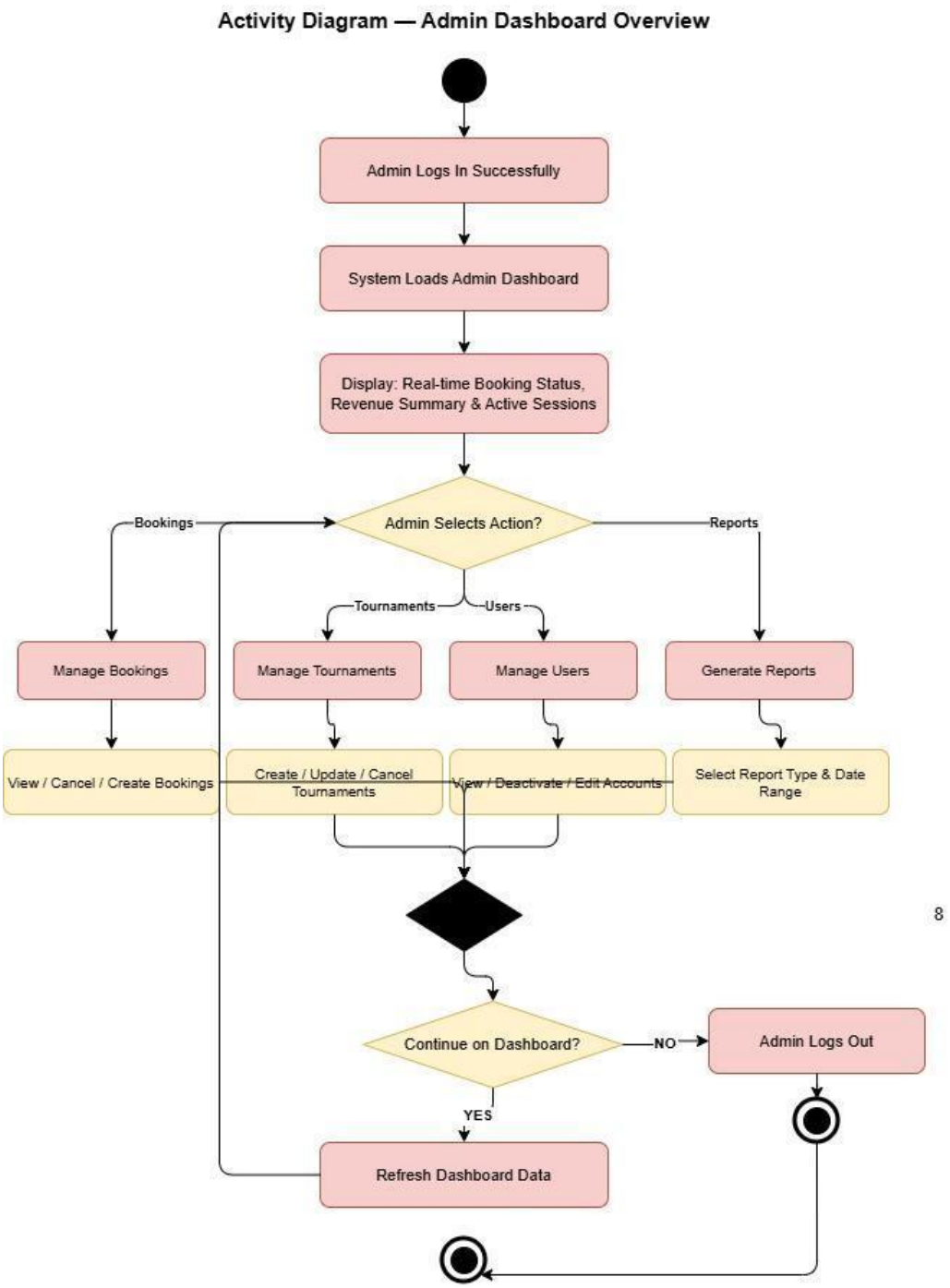


Figure 9: Activity Diagram – Admin Dashboard Overview

7.9 Booking Cancellation and Rescheduling

This activity diagram describes the process by which a customer cancels or reschedules a confirmed booking, including the 2-hour cutoff check, slot release, and loyalty point reversal.

Step	Activity	Actor	Decision / Outcome
1	Open My Bookings; select a CONFIRMED booking	Customer	–

Step	Activity	Actor	Decision / Outcome
2	Check cutoff	System	Decision: Start time more than 2 hours away? NO → Return TOO_LATE → END
3	Choose action	Customer	Cancel | Reschedule
4a	Cancel: set status CANCELLED; release slot	System	Broadcast slot_unlocked
5a	Reverse loyalty points; log REVERSED transaction; initiate refund	System	If reversal fails, queue compensating transaction
4b	Reschedule: select available slot within 30 days	Customer	–
5b	Move booking; release old slot; lock new slot	System	–
6	Notify customer (email / SMS)	System	Postcondition: Slot state and loyalty balance consistent

Table 44: Activity Steps – Booking Cancellation and Rescheduling

7.10 Session Review

This activity diagram describes the process by which a customer rates a completed gaming session.

Step	Activity	Actor	Decision / Outcome
1	Open My Bookings; choose Rate Session on a completed booking	Customer	–
2	Check eligibility	System	Decision: Completed, owned, and not yet reviewed? NO → Show reason → END
3	Select 1–5 stars and optional comment; submit	Customer	–
4	Save review; update station average rating	System	Postcondition: Review visible in My Reviews and admin view

Table 45: Activity Steps – Session Review

Chapter 8 External Interface Requirements

8.1 User Interface Requirements

A clean, intuitive, and responsive web-based user interface shall be provided by the system. The following requirements apply:

- The interface shall be accessible on desktop, tablet, and mobile devices through responsive design (minimum viewport width of 320px) and shall conform to WCAG 2.1 Level AA.
- The customer-facing interface shall include pages for: Home, Slot Booking (with station category filter, rating badges, and live "Holding" indicators), Tournaments (with ticker banner), My Bookings, My Reviews, Loyalty Rewards, Profile, and Payment History.
- The admin interface shall include a dashboard with real-time booking status, revenue summary, station and game management, tournament management, user management, a notification centre, and reporting tools.
- Error messages shall be displayed inline, written in plain English, and accompanied by suggested corrective actions; policy errors such as TOO_LATE shall map to specific messages.
- A consistent dark gaming colour scheme, typography, and iconography aligned with the Gaming Castle brand shall be used by the interface.

8.2 Hardware Interface Requirements

The Gaming Castle system is a web-based application and does not require direct integration with specialised hardware. However, the following hardware dependencies apply:

- The hosting infrastructure shall provide the container resources defined in the SDS (minimum 0.5 vCPU and 512 MB RAM per service instance, scalable under load).
- A modern web browser and stable internet connectivity are required at administrator terminals.
- No direct hardware-level integration with PS5 consoles, PCs, or VR headsets is required; station allocation is managed logically within the system.

8.3 Software Interface Requirements

- Payment Gateway: PayHere shall be integrated for LKR payments, including MD5 merchant signature generation and an IPN webhook endpoint.
- Email Service: An SMTP email service (JavaMail) shall be integrated for booking confirmations, tournament notifications, and verification codes.
- SMS Notifications: Twilio shall be integrated for SMS confirmations and one-time verification codes.

- Real-Time Server: A Node.js Socket.IO server shall provide live slot, notification, and tournament events, with an HTTP event bridge from backend services.
- Database: PostgreSQL 16 shall be used for persistent data storage, with a separate database per service.
- Service Registry and Gateway: Netflix Eureka and Spring Cloud Gateway shall provide service discovery and routing.

8.4 Communications Interface Requirements

- All client-server communications shall use HTTPS over TLS 1.2 or higher.
- RESTful APIs shall be used for communication between the frontend and backend and between services, with data exchanged in JSON format.
- WebSocket (Socket.IO) connections shall carry real-time slot lock, unlock, booked, notification, and tournament events.
- The PayHere IPN endpoint shall be publicly reachable, exempt from JWT authentication, and protected by signature validation.
- The system shall operate over standard TCP/IP networks and shall not require any proprietary network protocols.

8.5 System Architecture

The Gaming Castle system shall be designed using a microservices-based architecture to improve scalability, maintainability, and flexibility.

- Six Spring Boot services shall be implemented: User, Booking (including stations, games, reviews, and Saga orchestration), Payment, Tournament, Loyalty, and Notification.
- A Node.js Socket.IO real-time server, a Spring Cloud API Gateway, and a Eureka service registry complete the backend.
- Each service shall own a separate PostgreSQL database; no cross-database queries shall be made, and services shall communicate through REST clients and real-time events.
- Inter-service REST clients shall be protected by circuit breakers, and multi-service transactions shall be compensated using the Saga pattern.
- Docker containerisation shall be used to package each component, ensuring consistent deployment across development, testing, and production environments.
- The API Gateway shall route requests, validate tokens, and enforce rate limiting on authentication routes.

This architecture provides the following benefits: independent deployment of services, improved system scalability, easier maintenance and debugging, and better fault isolation.

Chapter 9 Data Requirements

The following key data entities are required for the Gaming Castle system, grouped by owning service. References across services use identifiers only; no foreign keys cross database boundaries.

Service	Entity	Key Attributes	Description
User	User	user_id, name, email, phone, password_hash, role, avatar_url, is_active, failed_login_attempts, lockout_until	Customer and admin account information.
User	VerificationCode	id, user_id, code, purpose (PASSWORD_RESET, REACTIVATION), expires_at, used	One-time codes for password reset and reactivation.
Booking	GameStation	id, station_code, name, type (CONSOLE, PC, VR), hourly_rate, active, image_url	Bookable stations with per-type pricing.
Booking	Game	id, title, genre, cover_url, platforms, active, featured	Game catalogue.
Booking	StationGame	station_id, game_id	Many-to-many station and game mapping.
Booking	Booking	id, booking_reference, user_id, station_id, game_id, booking_date, start_time, end_time, total_amount, discount_amount, final_amount, status, booking_type	Online and walk-in bookings.
Booking	Review	id, booking_id (unique), user_id, station_id, rating (1–5), comment, created_at, updated_at	Session reviews.
Booking	CompensatingTransaction	id, booking_id, transaction_type, status, payload, retry_count	Saga compensation audit.
Payment	Payment	id, booking_id, user_id, amount, discount_amount, method (PAYHERE, CARD, CASH), status, transaction_reference, gateway_order_id, payhere_payment_id, loyalty_points_used	Payment records; no card data stored.
Loyalty	LoyaltyAccount	id, user_id (unique), balance_points, tier (BRONZE, SILVER, GOLD)	Per-customer balance and tier.
Loyalty	LoyaltyTransaction	id, account_id, booking_id, transaction_type (EARNED, REDEEMED, REVERSED), points, balance_after, created_at	Loyalty audit trail.
Tournament	Tournament	id, name, game_id, date, entry_fee, max_participants, participant_count, status, image_url, reminder_sent	Tournament events.

Service	Entity	Key Attributes	Description
Tournament	TournamentRegistration	id, tournament_id, user_id, user_name, user_email, payment_id, registered_at	Registrations (participant details copied at registration).

Table 46: Data Requirements – Key Entities

All personal data (names, emails, phone numbers) shall be stored in compliance with applicable data protection regulations. Customer passwords shall be stored exclusively as bcrypt hashes, and card details shall never be stored.

Chapter 10 Alternative Solutions Considered

Alternative	Description	Reason for Rejection
WhatsApp / Phone-based Booking	Continuation with manual booking via phone calls or WhatsApp messages.	Does not scale; prone to human error; no automated payment or loyalty tracking; rejected in favour of the proposed web system.
Third-party Booking Platform	Adoption of an existing venue booking platform (e.g., Booksy, SimplyBook.me).	Generic platforms do not support gaming-specific features (tournament management, per-station booking, loyalty points); high recurring subscription costs.
Mobile Application (Native)	Development of a native iOS/Android mobile application instead of a web application.	Higher development cost and time; separate codebases are required for iOS and Android; a responsive web application achieves the same goals with a single codebase.
Spreadsheet-based System	Use of Google Sheets or Microsoft Excel for booking and payment tracking.	Not suitable for real-time multi-user access; lacks payment integration and automated notifications; not scalable.
Database row-locking for slot holds	Hold slots through pessimistic locks or hold rows in the relational database.	Creates write contention and lock waits on busy slots; replaced by an in-memory TTL lock on the real-time server backed by a database unique constraint.
Distributed (two-phase) transactions	Atomic commit across booking, payment, and loyalty databases.	Couples services and blocks on failures; the Saga pattern with compensating transactions provides eventual consistency with database-per-service.
Shared database for all services	One schema accessed by every service.	Tight coupling and shared failure; rejected in favour of database-per-service.

Table 47: Alternative Solutions Considered

Chapter 11 Feasibility Study

11.1 Technical Feasibility

The proposed system is technically feasible and has been implemented. The required skills for building a web application using Angular 19 (frontend), Spring Boot 3 with Java 17 (backend), Node.js with Socket.IO (real-time), and PostgreSQL 16 are available within the development team. PayHere provides documented integration for Sri Lankan payments, and Resilience4j, Eureka, and Spring Cloud Gateway are mature, well-documented components. No novel or unproven technologies are required.

The use of Docker and microservices architecture further strengthens the technical feasibility of the system. Docker ensures a consistent runtime environment across different stages of development, while microservices enable modular development and scalability. The development team has sufficient knowledge to implement and manage containerised services within the project scope.

An automated tournament management module is implemented using rule-based logic. Given the scale of the system, a lightweight approach is sufficient to generate tournament brackets and manage match outcomes without requiring complex infrastructure.

11.2 Economic Feasibility

The system is developed using open-source frameworks and tools at minimal licensing cost. Cost-effective deployment options are offered by cloud hosting providers (AWS Free Tier). The expected development cost is limited to the time investment of the four-member team within the academic project context. For Gaming Castle, a return on investment is projected to be generated by reducing manual overhead, increasing utilisation across PS5, PC, and VR stations, and attracting a broader customer base through online accessibility, loyalty rewards, and tournament events. PayHere transaction fees should be reflected in pricing.

11.3 Operational Feasibility

The system is operationally feasible. The administrator interface is designed to be intuitive and requires minimal training. The customer-facing interface follows familiar web conventions for booking and payment. Real-time slot indicators reduce booking conflicts, and reviews give management direct feedback on stations and equipment. Internet connectivity is available at the Gaming Castle premises, and web-based services are routinely used by the target users. Customer engagement and retention are expected to be increased through the loyalty programme and tournament features.

11.4 Schedule Feasibility

The project is planned to be completed within the academic semester. Four members with defined responsibilities constitute the team. The development is structured into sprints covering requirements analysis, system design, frontend development, backend development,

integration testing, and final deployment. The implemented scope and the automated test suite show that the schedule is achievable within the allocated timeframe.

Chapter 12 Risk Analysis

The following Risk Register documents identified project risks, their likelihood, impact, overall risk level, and proposed mitigation strategies.

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R-01	Payment gateway integration failure or delays	Technical	Medium	High	High	The PayHere sandbox shall be used for testing; a mock payment module shall be maintained; IPN handling shall be validated with automated tests.
R-02	Scope creep due to evolving client requirements	Requirements	High	Medium	High	A formal change request process shall be established; client sign-off on requirements shall be obtained before development.
R-03	Team member unavailability due to illness or academic workload	Resource	Medium	Medium	Medium	All work shall be thoroughly documented; knowledge shall be distributed across the team; task backups shall be maintained.
R-04	Security vulnerabilities in the web application	Technical	Medium	High	High	An OWASP Top 10 assessment shall be conducted; HTTPS, input validation, and gateway rate limiting shall be enforced; peer code reviews shall be performed.
R-05	Project timeline overrun	Schedule	Medium	High	High	Agile sprint planning shall be used; weekly progress reviews shall be conducted; core features shall be prioritised for MVP delivery.
R-06	Low user adoption post-deployment	External	Low	Medium	Low	The client and target users shall be involved in UAT; a user guide and onboarding support shall be provided.
R-07	Third-party API deprecation or changes	Technical	Low	High	Medium	Stable, well-maintained APIs shall be selected; abstraction layers (payment, SMS, email adapters) shall be implemented to facilitate API replacement.

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R-08	Data loss due to server failure	Technical	Low	High	Medium	Automated daily backups of every service database with off-site storage shall be implemented; recovery procedures shall be tested.
R-09	Inconsistent data across service databases after partial failures	Technical	Medium	High	High	Saga compensating transactions with an audit table and retry count shall be used; circuit breakers shall protect inter-service clients; failure scenarios shall be covered by automated tests.
R-10	Loss of in-memory slot holds if the real-time server restarts, or inability to scale it horizontally	Technical	Medium	Medium	Medium	The database unique constraint remains the authoritative guard; holds expire in 5 minutes; clients resynchronise on reconnect; a shared store shall be evaluated for multi-instance deployment.
R-11	Missed or delayed PayHere IPN leaving a payment unconfirmed	Technical	Medium	High	High	The IPN handler shall be idempotent; pending payments shall be reconciled; hold expiry and customer notification shall be handled.
R-12	Brute-force and credential-stuffing attacks on authentication	Security	Medium	High	High	Token-bucket rate limiting (5 req/s, burst 10), account lockout after 5 failures, and bcrypt hashing shall be applied.

Table 48: Risk Register

Chapter 13 References

[1] IEEE, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Std 830-1998, 1998.

[2] ISO/IEC, "Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE) – System and software quality models," ISO/IEC 25010:2011, 2011.

[3] OWASP Foundation, "OWASP Top Ten," 2021. [Online]. Available: https://owasp.org/www-project-top-ten/. (accessed Mar. 11, 2026).

[4] W3C, "Web Content Accessibility Guidelines (WCAG) 2.1," 2018. [Online]. Available: https://www.w3.org/TR/WCAG21/. (accessed Mar. 11, 2026).

[5] PayHere, "PayHere Developer Documentation (Checkout API and Payment Notification)," [Online]. Available: https://support.payhere.lk/. (access date to be added by the authors).

[6] C. Richardson, *Microservices Patterns*. Manning Publications, 2018.

[7] H. Garcia-Molina and K. Salem, "Sagas," in *Proc. ACM SIGMOD Int. Conf. on Management of Data*, 1987.

[8] Resilience4j, "Resilience4j Documentation," [Online]. Available: https://resilience4j.readme.io/. (access date to be added by the authors).

[9] Socket.IO, "Socket.IO Documentation," [Online]. Available: https://socket.io/docs/. (access date to be added by the authors).

Client Requirement Sign-Off

Client Representative Sign-Off

Name: _____ Signature: _____ Date: _____

Project Supervisor Sign-Off

Name: _____ Signature: _____ Date: _____